Complex Numbers: Cartesian Form & Algebra

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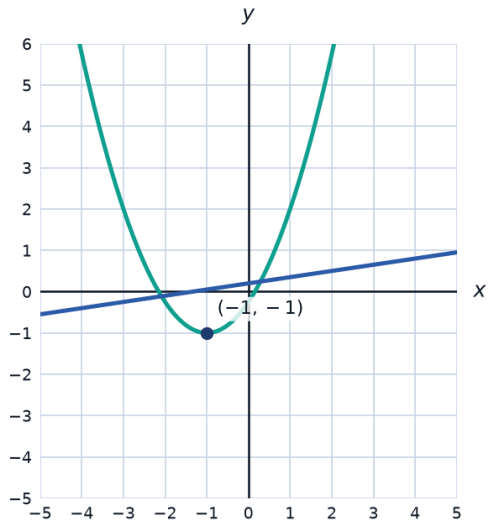
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

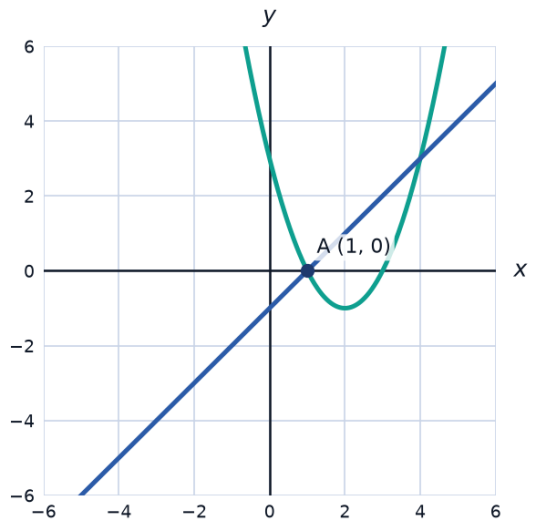
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the complex number w such that $w = x + yi$, where x and y are real numbers. Given that $w^3 = -64i$ and the imaginary part of w is positive, find the value of $x^2 + y^2$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

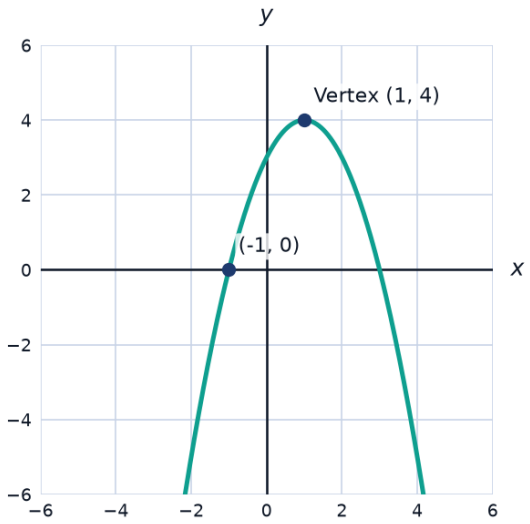
The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.



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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of b.



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Show that the sum of these five roots equals 0.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the complex number z such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that a is a real number, find the possible values of a.

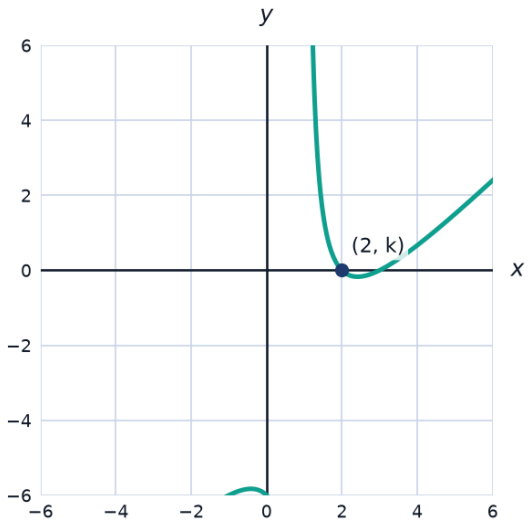
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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

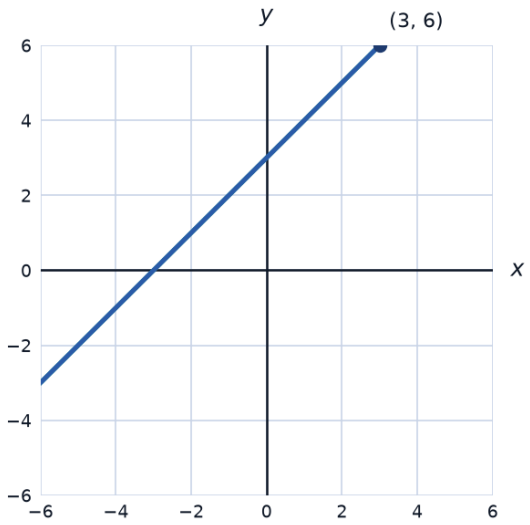
11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.



12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

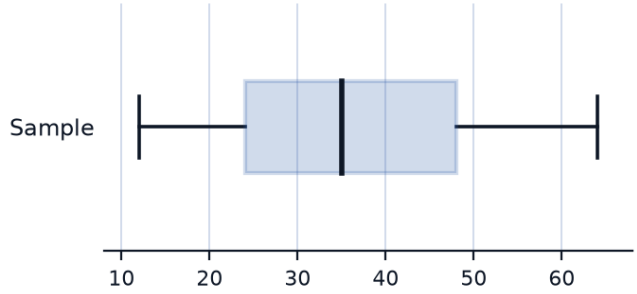
Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of k. The graph of $y = f(x)$ is shown below.



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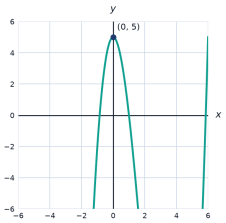
13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.



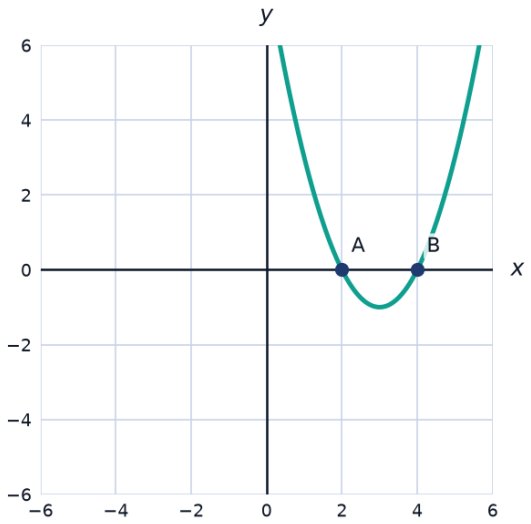
(4, -27)

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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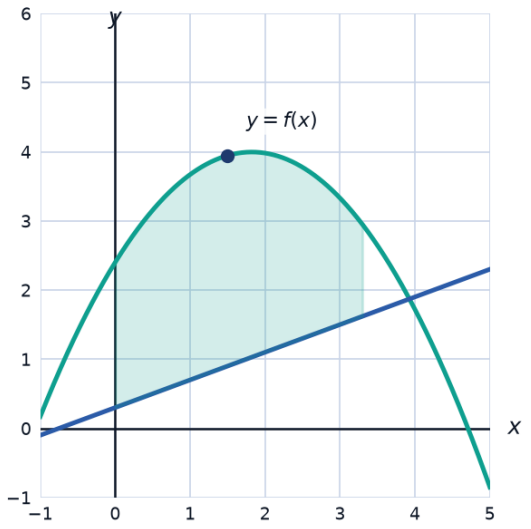
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?



16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Proof by Mathematical Induction

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.04 Partial Fractions

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

a. $5 + 14i$

b. $-3 - 8i$

c. $5 - 14i$

d. $-3 + 14i$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $\frac{4+2i}{1-i}$ in Cartesian form $a + bi$.

a. $1 + 3i$

b. $3 + 3i$

c. $1 - 3i$

d. $2 + 2i$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = x + iy$, what is $z \cdot z^*$ (where z^* is the complex conjugate)?

a. $x^2 + y^2$

b. $x^2 - y^2$

c. $x^2 + 2ixy - y^2$

d. $2x$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

a. $|z| = 2, \text{Arg}(z) = \frac{5\pi}{6}$

b. $|z| = 2, \text{Arg}(z) = -\frac{\pi}{6}$

c. $|z| = 4, \text{Arg}(z) = \frac{2\pi}{3}$

d. $|z| = 2, \text{Arg}(z) = \frac{7\pi}{6}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus of $\frac{3+4i}{1-2i}$?

a. $\sqrt{5}$

b. $\frac{5}{3}$

c. 5

d. $\frac{25}{5} = 5$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the principal argument of $z = -5i$.

a. $-\frac{\pi}{2}$

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{2}$

d. π

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the two square roots of $-8 - 6i$ in Cartesian form.

a. $\pm(1 - 3i)$

b. $\pm(3 - i)$

c. $\pm(1 + 3i)$

d. $\pm(2 - 3i)$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the quadratic equation $z^2 - 4z + 13 = 0$ over $\mathbb{C}$.

a. $z = 2 \pm 3i$

b. $z = 4 \pm 6i$

c. $z = -2 \pm 3i$

d. $z = 2 \pm 9i$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Describe the locus of points in the complex plane satisfying $|z - 3 + 2i| = 5$.

a. A circle with center $(3, -2)$ and radius 5

b. A circle with center $(-3, 2)$ and radius 5

c. A straight line perpendicular to $3 - 2i$

d. An ellipse with focal distance 5

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

What geometric locus is defined by $|z - 1| = |z - 3i|$?

a. The perpendicular bisector of the segment connecting $(1, 0)$ and $(0, 3)$

b. A circle passing through $(1, 0)$ and $(0, 3)$

c. A line segment connecting $(1, 0)$ and $(0, 3)$

d. A parabola opening upward

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the Cartesian equation of the locus $\text{Arg}(z - 2) = \frac{\pi}{4}$.

a. The half-line $y = x - 2$ for $x > 2$

b. The full line $y = x - 2$

c. The circle $(x - 2)^2 + y^2 = 1$

d. The half-line $y = -x + 2$ for $x < 2$

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct complex nth roots does the equation $z^7 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 7 roots, $\frac{360}{7}^\circ$ apart

b. 8 roots, $\frac{360}{7}^\circ$ apart

c. 7 roots, 45° apart

d. 6 roots, $\frac{360}{7}^\circ$ apart

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $z = x + yi$ be a complex number. The modulus of z, denoted by $|z|$, is given by $|z| = \sqrt{x^2 + y^2}$. Given that $|z| = \sqrt{5}$ and the real part of z is 2, find the possible values for the imaginary part of z.

a. $\pm\sqrt{3}$

b. ± 3

c. $\pm\sqrt{1}$

d. ± 1

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. 5

b. 7

c. 25

d. $\sqrt{7}$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = 3 + ki$, where k is a real parameter. The modulus of z is given by $|z| = \sqrt{13}$. Find the positive value of k.

a. 1

b. 2

c. 3

d. 4

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the complex number $z = -3 + 4i$. Find the modulus of z, denoted by $|z|$.

a. 5

b. 7

c. 25

d. 16

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A cubic equation with real coefficients $x^3 - 10x^2 + 37x - 52 = 0$ has $3 + 2i$ as one root. Find the real root.

a. 2

b. 4

c. 3

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

For which value of m does the system $x + my = 3,\ 2x + 4y = 6$ have infinitely many solutions?

a. 2

b. 4

c. 1

d. 3

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.04 Partial Fractions

How many solutions does the system $x + y = 4,\ 2x + 2y = 8$ have?

a. Infinitely many

b. No solution

c. Exactly one

d. Exactly two

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find all cube roots of unity.

a. $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

a. $|z^6| = 64,\ \arg(z^6) = 60^\circ$

b. $|z^6| = 12,\ \arg(z^6) = 0^\circ$

c. $|z^6| = 64,\ \arg(z^6) = 90^\circ$

d. $|z^6| = 64,\ \arg(z^6) = 0^\circ$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

In the expansion of $(1 + x)^7$, find the ratio of the coefficient of x^4 to the coefficient of x^3.

a. 1

b. 2

c. 3

d. 6

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find $\bar{z}$ (the complex conjugate) if $z = -3 + 9i$.

a. $-3 - 9i$

b. $-3 + 9i$

c. $3 - 9i$

d. $3 + 9i$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the coefficient of x^5 in the expansion of $(1 + 2x)^7$.

a. 519

b. 780

c. 787

d. 672

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Which value of k makes the system $x + 2y = 5,\ 2x + 4y = k$ have infinitely many solutions?

a. 10

b. 5

c. 20

d. 15

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